Let's Start

Efficiency Frontier

(VGG16 vs MobilenetV2)

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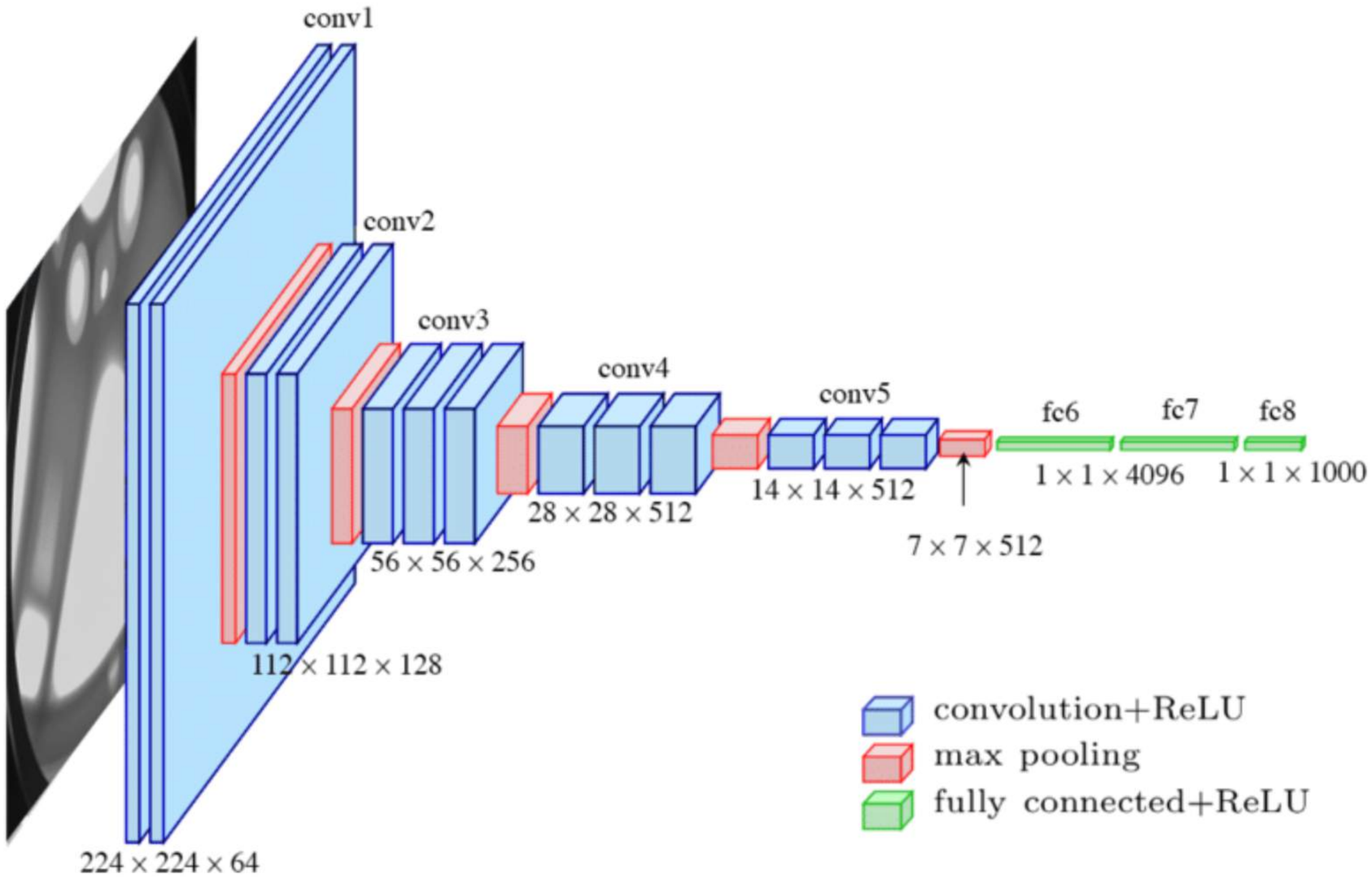
Problem statement

High-performance Deep Learning architectures often carry extreme computational redundancy, making them unsustainable for large-scale production

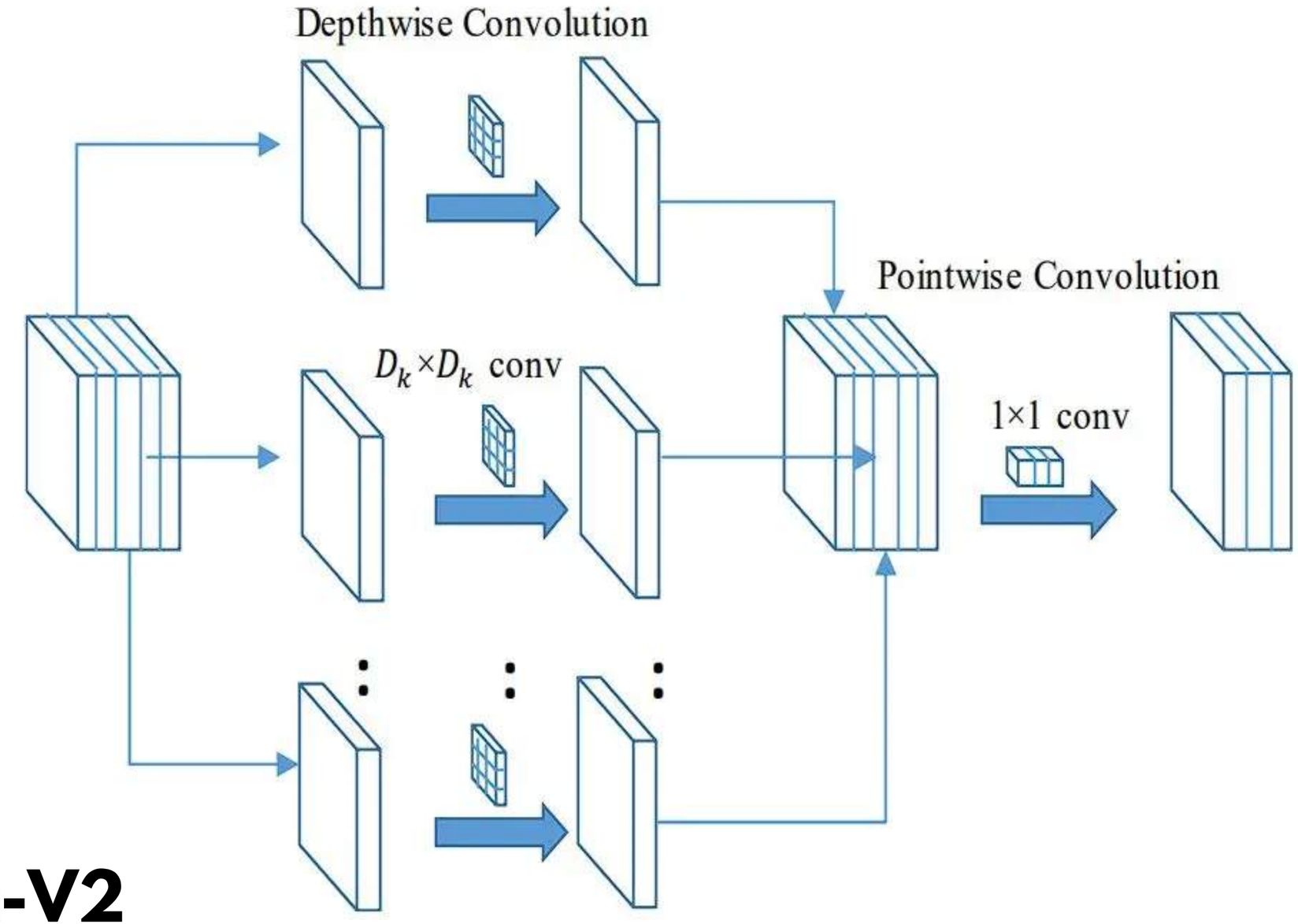
Objective: To identify the "**Efficiency Frontier**" — proving that architectural optimization can deliver superior accuracy while reducing the mathematical workload

Models



VGG-16

Models



MobileNet-V2

Resources usage

	VGG 16	Mobilenet -V2
FLOPs	134.302M	2.237M
Parameters	15.466 Billion	326.220 Million
Model Size	512.32 MB	8.53 MB

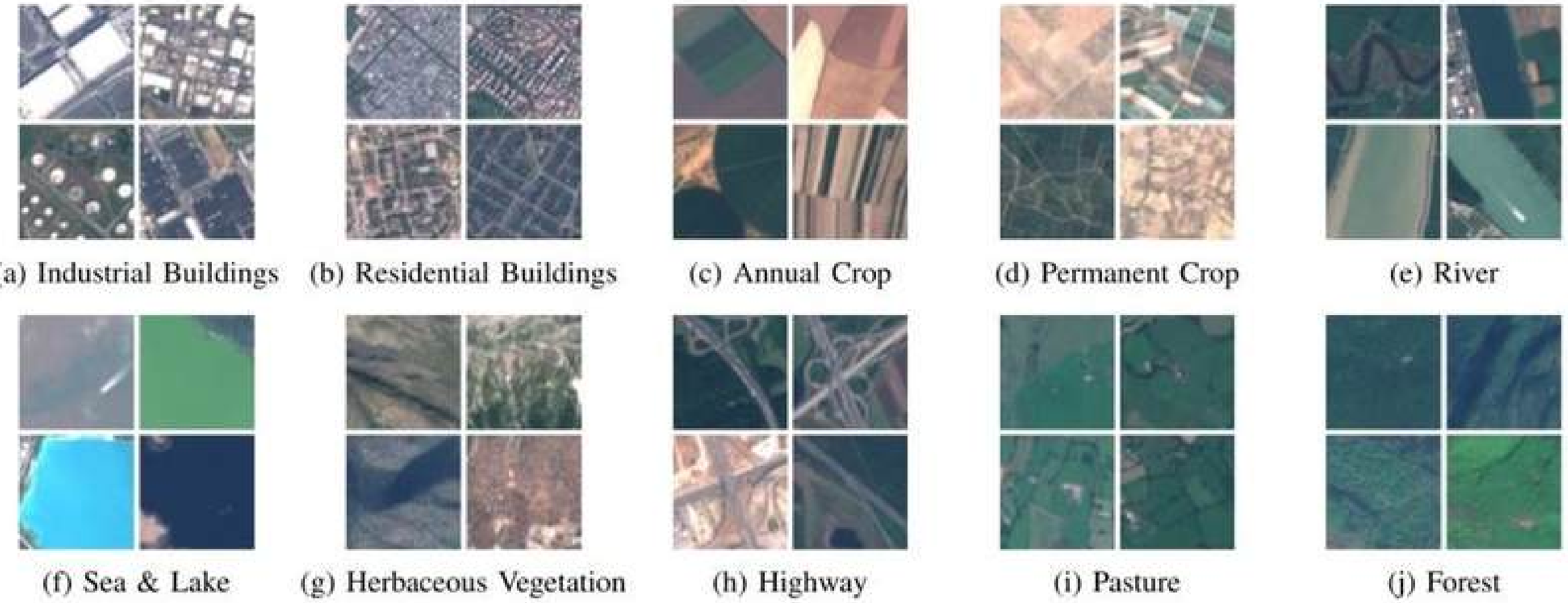
VGG16 is **47.4x** more computationally expensive than MobileNetV2

Approach

- Dataset and pre-processing
- Transfer-Learning
- Fine-Tuning Models
- Key Findings
- Visualisation

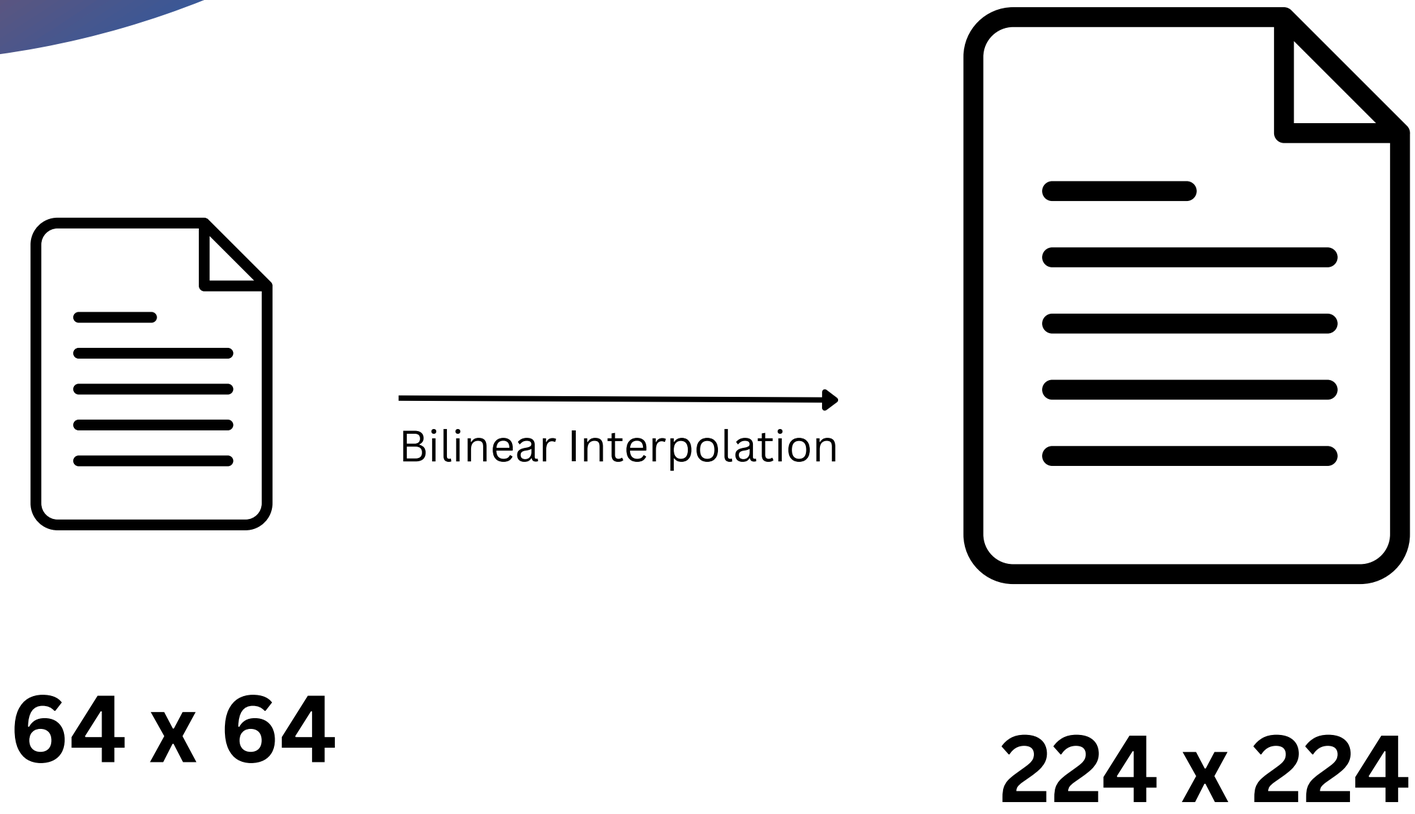
Dataset

- **Euro-Sat Dataset**



Pre-Processing

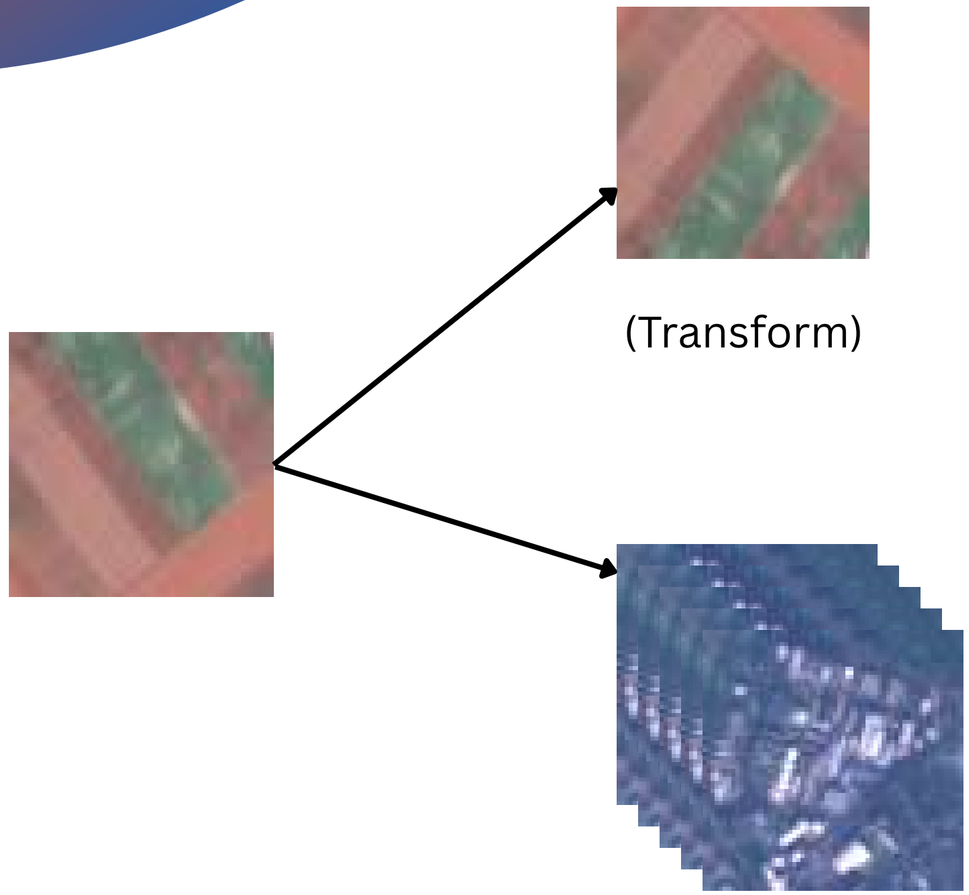
Image Re-Sizing



- A resampling technique used to upscale the original 64×64 EuroSAT images to the 224×224 size required by VGG16 and MobileNetV2.

- It estimates the color of a "new" pixel by taking a weighted average of the 4 nearest known pixels (a 2×2 neighborhood).

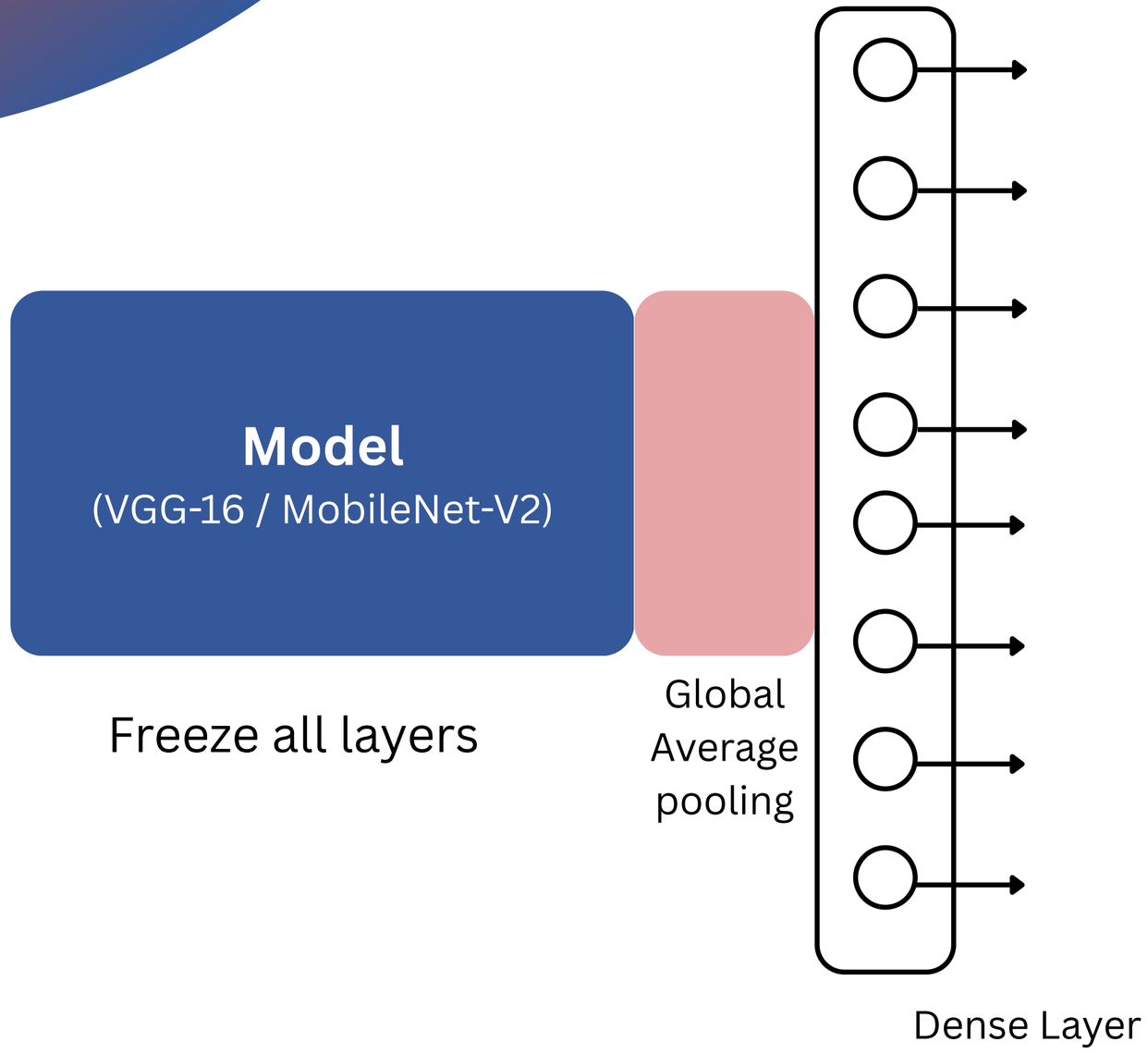
Data Augmentation & Batch Sizing



- Prevents Overfitting: Forces the model to learn general patterns rather than memorizing specific training samples.
- Rotational Invariance: Simulates diverse satellite flight paths using horizontal flips and 15 degree rotations.
- Optimized Batch Processing: Utilizes a Batch Size of 32 to balance computational efficiency with stable gradient updates during training.
- Standardized Normalization: Applies ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] to align input data with the pretrained model weights.

Transfer learning

Transfer Learning / Warmup



Frozen Feature Extraction: All initial layers were frozen to prevent the destruction of pre-trained weights during the early stages of training.

Modified Classification Head: Replaced the original 1000-class head with a custom architecture:

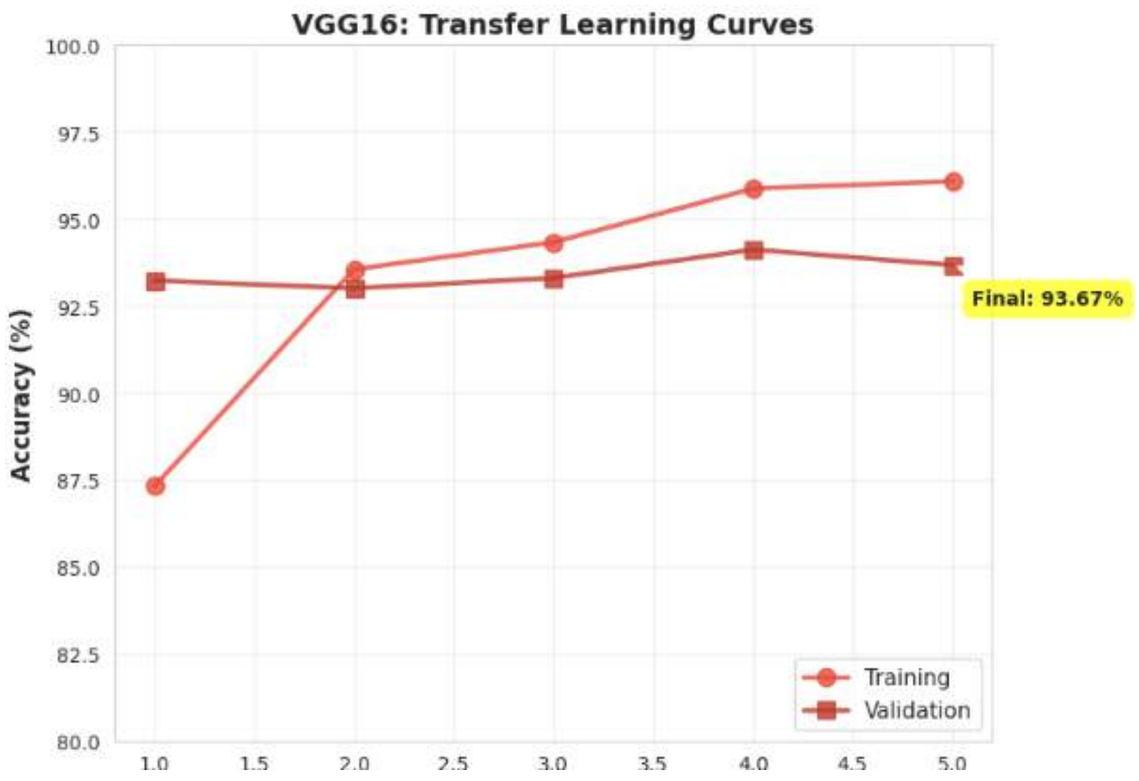
- Global Average Pooling: To reduce spatial dimensions .
- Dense Layer (Softmax): A 10-node layer mapping features to the 10 classes .

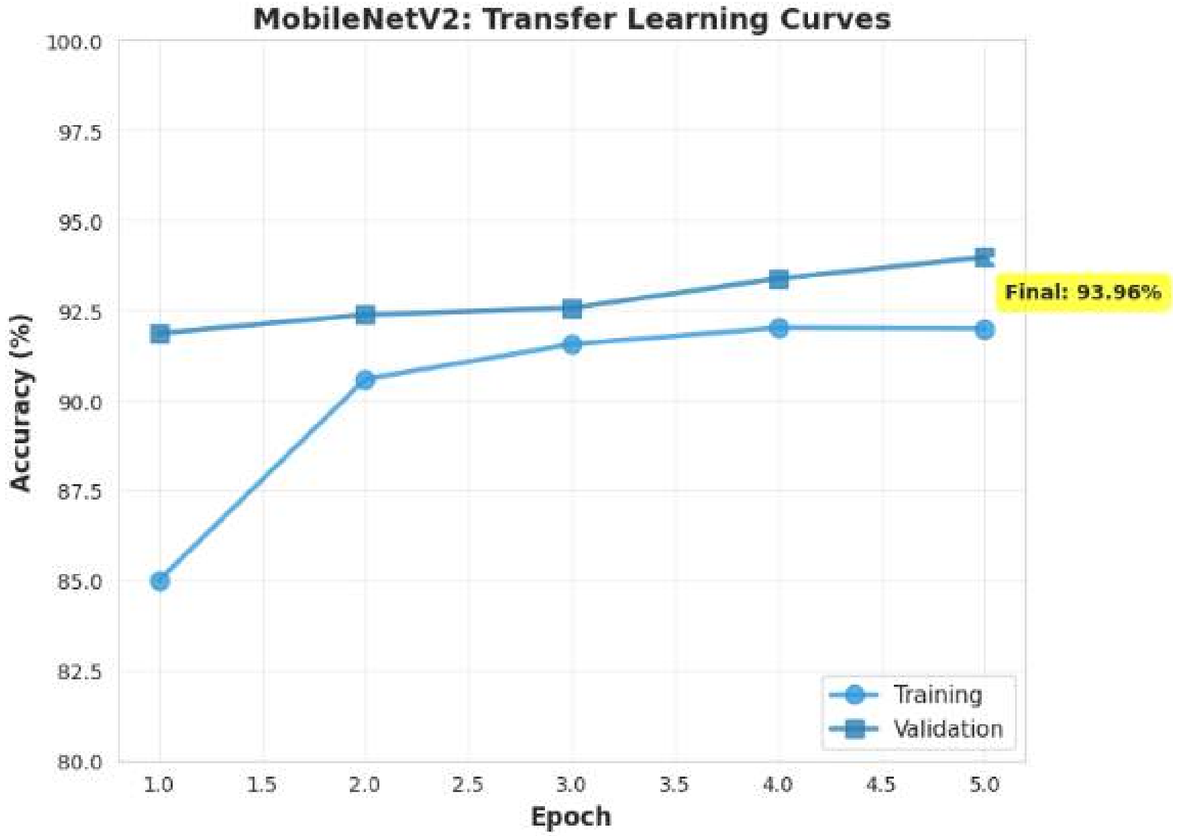
Ran 5 **Epochs** Each

Results

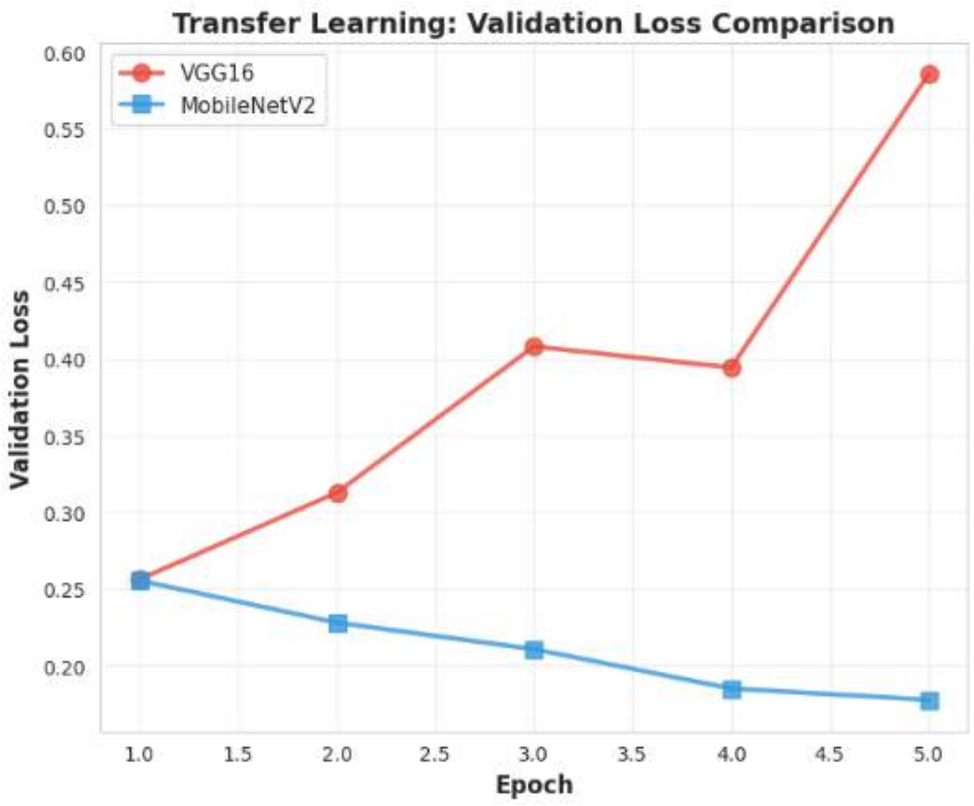
	VGG 16	Mobilenet -V2
Total params	134.302M	2.237M
Trainable	16.78 M	0.01M
%Trained	12.50%	0.57%
New FLOPs	15.47 billion	0.33 billion
Accuracy	93.67%	93.96%

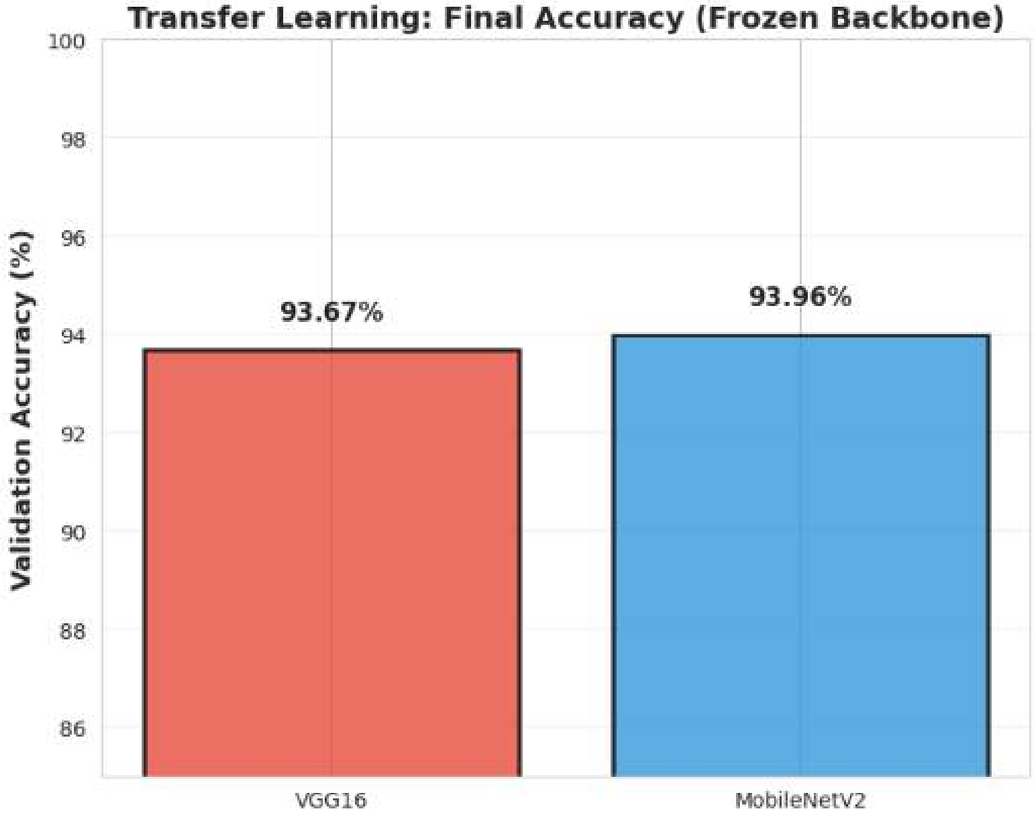
Results


VGG16: Transfer Learning Curves
Accuracy (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Final: 93.67%
Training
Validation


MobileNetV2: Transfer Learning Curves
Accuracy (%)
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
80.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Epoch
Final: 93.96%
Training
Validation

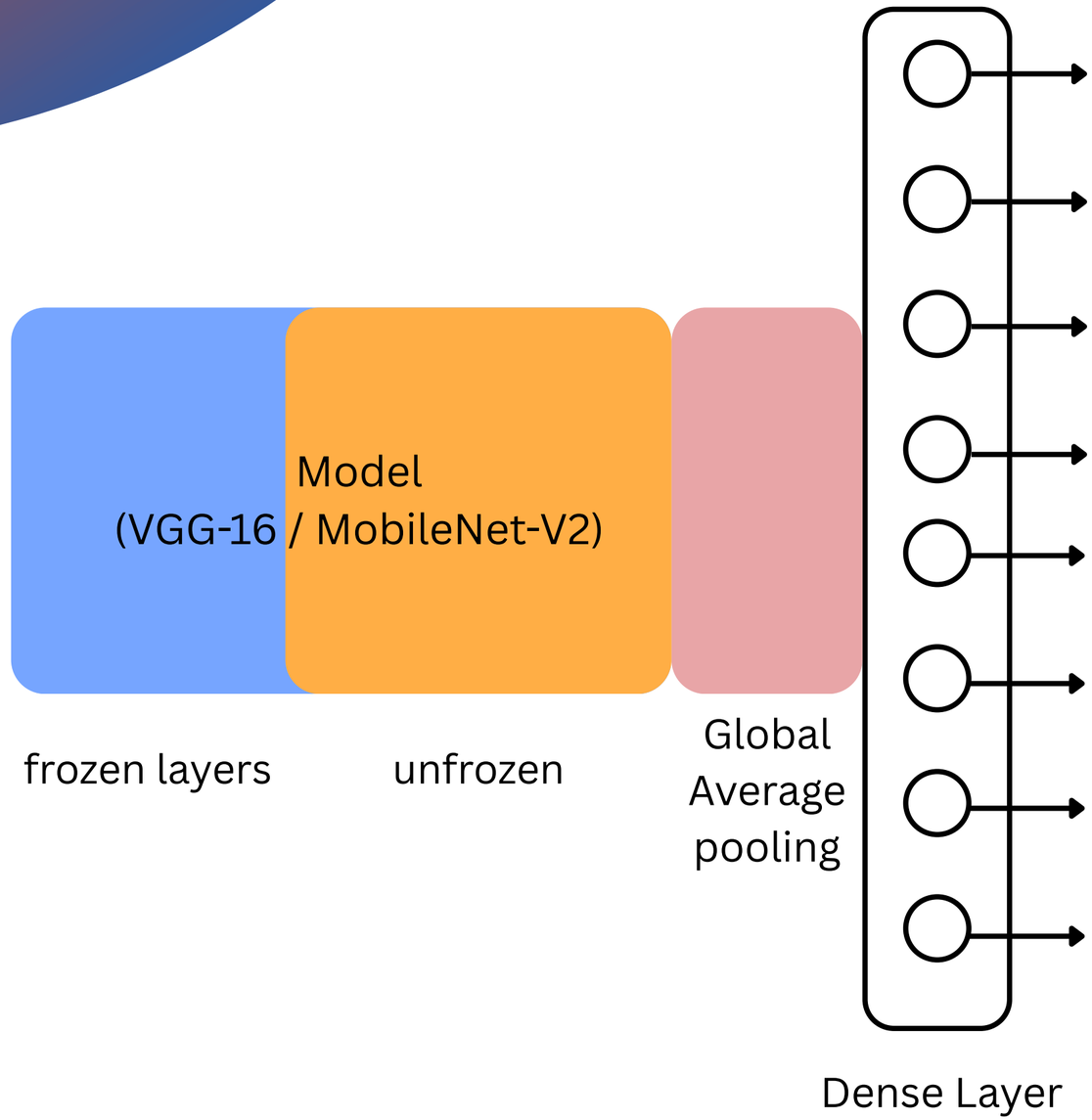
Results


Transfer Learning: Validation Loss Comparison
VGG16
MobileNetV2
Validation Loss
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Epoch


Transfer Learning: Final Accuracy (Frozen Backbone)
Validation Accuracy (%)
100
98
96
94
92
90
88
86
93.67%
93.96%
VGG16
MobileNetV2

Fine-Tuning

Fine-Tuning



- **Targeted Layer Unfreezing:** from a frozen backbone to unfreezing the final architectural blocks (Block 5 for VGG16; last 3 residual blocks for MobileNetV2)

- **Weight Refinement**: Allowed the model to update its high-level feature detectors while preserving low-level detectors (edges/colors) from ImageNet.

- **Learning Rate Optimization**: significantly lower learning rate(1e-5) during this phase to maintain stability and prevent "**catastrophic forgetting**" of pre-trained knowledge.

- Also used Early stopping to overcome overfitting

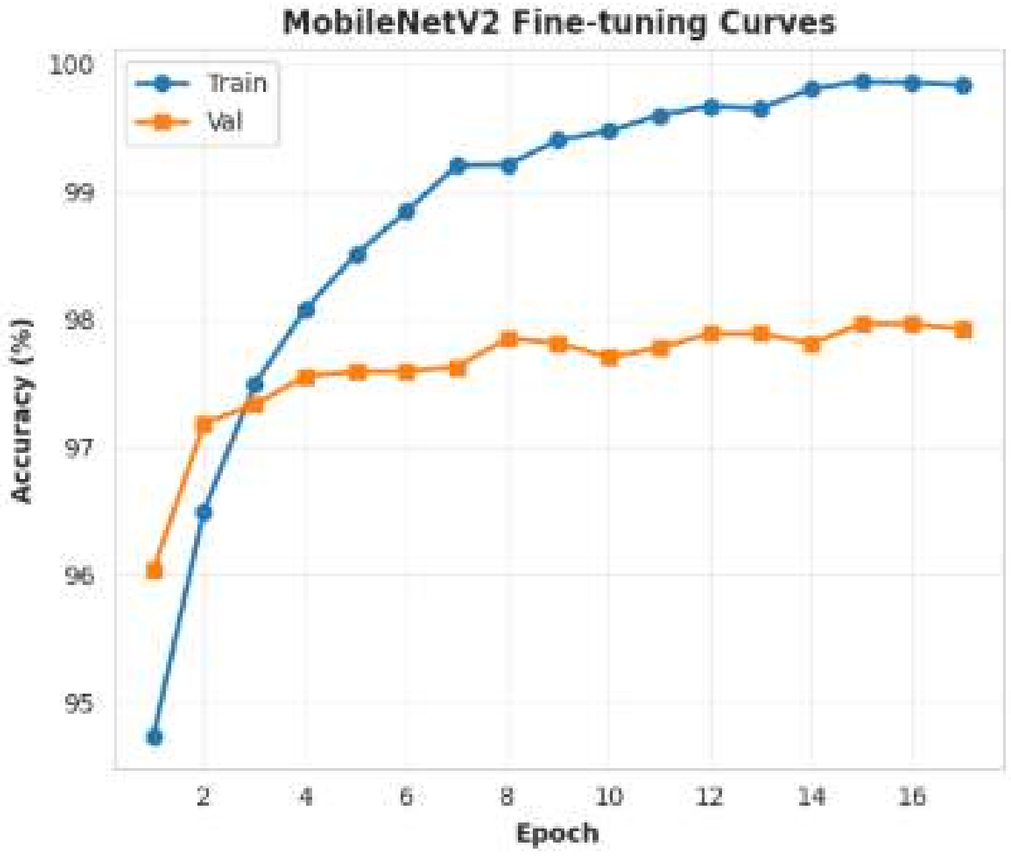
Benchmarking Results

	VGG 16	Mobilenet -V2
Mean Latency	4.86 ms/image	0.85 ms/image
Throughput	205.86 images/sec	1179.7 images/sec
Peak Memory	6229.98 MB	1267.01 MB
New FLOPs	15.47 billion	0.33 billion
Accuracy	95.93%	97.96%

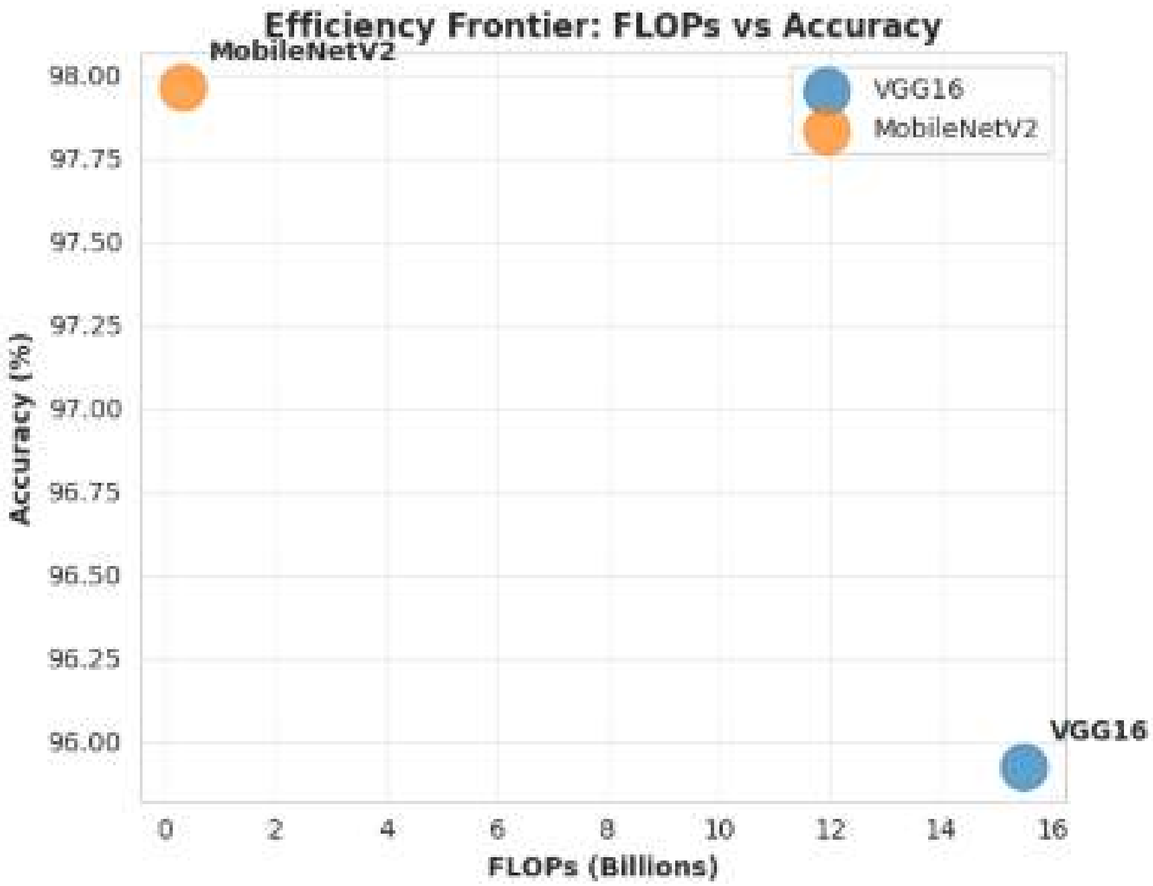
MobileNetV2 is **5.73x** faster than VGG16

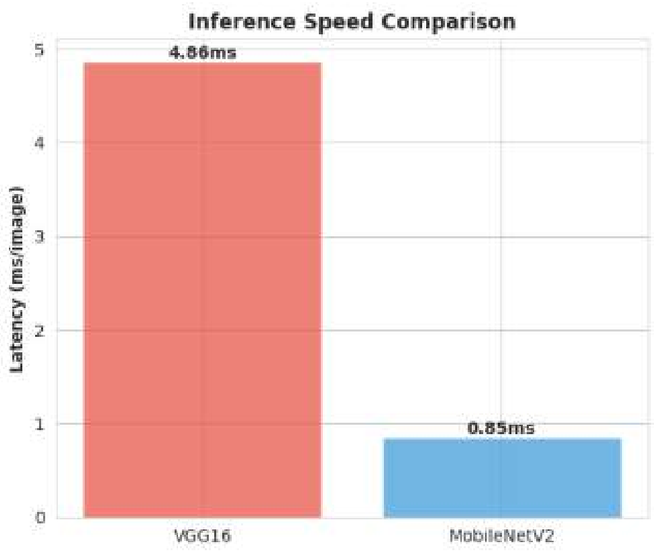
Results

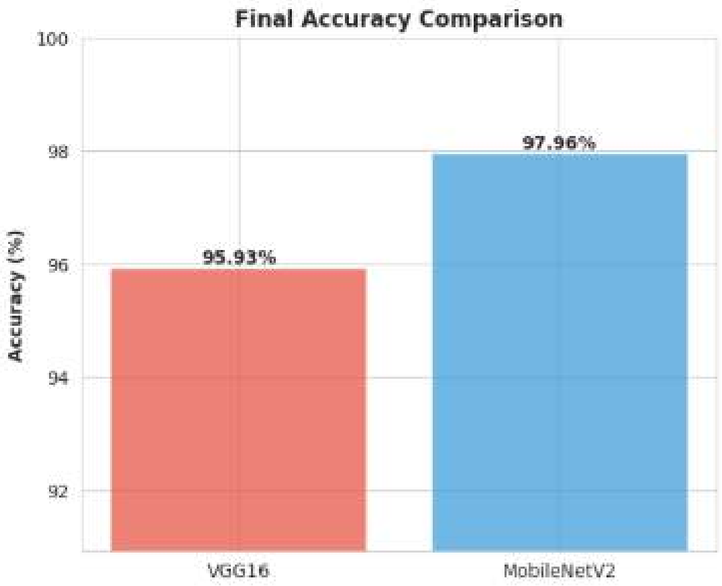
VGG16 Fine-tuning Curves
Train
Val
Accuracy (%)
99
98
97
96
95
2
4
6
8
10
Epoch


MobileNetV2 Fine-tuning Curves
Train
Val
Accuracy (%)
100
99
98
97
96
95
2
4
6
8
10
12
14
16
Epoch

Results


Efficiency Frontier: FLOPs vs Accuracy
MobileNetV2
VGG16
MobileNetV2
VGG16
Accuracy (%)
98.00
97.75
97.50
97.25
97.00
96.75
96.50
96.25
96.00
0
2
4
6
8
10
12
14
16
FLOPs (Billions)


Inference Speed Comparison
4.86ms
0.85ms
Latency (ms/image)
5
4
3
2
1
0
VGG16
MobileNetV2


Final Accuracy Comparison
97.96%
95.93%
Accuracy (%)
100
98
96
94
92
VGG16
MobileNetV2

Scan Me!

https://aravind-1294-deep-learning-model-c9masn.streamlit.app

End

Thank you

Do you have any questions?